

Monthly Systems Health & Operations Report

Infrastructure Engineering Group • US-East Primary Availability Zone

STATUS: **DEGRADED**
SLA Warning Raised

REPORTING PERIOD
Jul 16 – Aug 15, 2026

GENERATED TIMESTAMP
Aug 15, 2026, 23:59 UTC

PRIMARY ENVIRONMENT
Production (Tier-1 Services)

CLUSTER ID
use1-prod-k8s-04

1. EXECUTIVE OVERVIEW

During the thirty-day window ending August 15, 2026, total platform throughput reached **42.8 million requests**. While global core availability maintained contractual baseline at **99.91%**, severe tail-latency degradation occurred in downstream data layer operations starting August 3, 2026. Unindexed queries introduced in release v2.14.0 caused connection queue exhaustion across the primary Postgres cluster, triggering p99 latency breaches.

MONTHLY AVAILABILITY

99.91%

Target SLA: $\geq 99.90\%$

TAIL LATENCY (P99)

412 ms

+62 ms above target ceiling

DB POOL SATURATION

78.4%

+8.4% threshold overrun

2. OPERATIONAL SLA METRICS BENCHMARK

System Metric	Baseline SLA Target	Measured August Value	Peak Observed	Compliance Status
Service Uptime (Global)	$\geq 99.90\%$	99.91%	99.99%	PASS
API Median Latency (p50)	< 120 ms	98 ms	145 ms	PASS
API Tail Latency (p99)	< 350 ms	412 ms	740 ms	BREACH
Error Rate (HTTP 5xx)	< 0.05%	0.072%	0.290%	WARN
Database Connection Pool	< 70.0%	78.4%	94.2%	WARN
Worker Pod Memory Headroom	> 25.0%	18.1%	11.4%	BREACH

3. CRITICAL INCIDENTS & REQUIRED CORRECTIVE ACTIONS

- INC-2026-08-03-A (Postgres Deadlock Spike):** Deployment v2.14.0 introduced unindexed joins on table transactions, starving connection threads for 14 minutes during peak lunch volume. Immediate rollback was applied.
- INC-2026-08-11-B (Worker Service Memory Leak):** A Node.js stream leakage in worker-service produced gradual OOM restarts across 6 replica pods over a 48-hour period.
- Required Patch Schedule:** Deploy missing composite index (`user_id`, `created_at`), add third read-replica to distribute analytics query load, and update runtime base images to patch container memory retention.